

On the Correctness of Real Parameterization, Modular Remark and Logarithmic Obstruction to Compatibility of Coefficient Symmetry

Statement of the Problem

The present text is an explanation for the current version of the article and the corresponding file `GlobalNormalization.v`. The filename is kept for backward compatibility of the GitHub repository; in version 16.4, its mathematical content corresponds to the logarithmic zero-gap obstruction to compatibility. The purpose of this explanation is to carefully separate five levels of reasoning:

real level,
integer level,
modular level,
coefficient symmetry compatibility level,
logarithmic zero-gap level of version 16.4.

At the real level, the parameterization

$$z = m^n + p^n, \quad x = m^n - p^n,$$

is considered, where the parameters

$$m, p$$

are not initially required to be integers. They can be treated as real parameters.

At the integer level, an additional question arises: when is such a real reconstruction compatible with the requirement

$$x, y, z \in \mathbb{N},$$

and, if necessary, with the requirement

$$m, p \in \mathbb{Z}?$$

This is exactly where the parity and exact power conditions appear.

At the modular level, a simple but important methodological remark is separately fixed: congruence modulo is not the same as equality in integers. Therefore, the existence of solutions for congruences modulo does not in itself constitute a counterexample to the integer formulation of Fermat's Last Theorem.

Finally, the central conditional step of version 16.4 is formulated via a comparison of two natural real normalizations of the same odd binomial core and via the logarithmic form of this comparison:

$$\text{core} = n l^n, \quad \text{core} = 2^{n-1} q^n.$$

Here

$$l > 0, \quad q > 0$$

are real residual factors. The comparison of these two representations yields the exact relation

$$l^n = q^n \frac{2^{n-1}}{n}.$$

Since

$$l > 0, \quad q > 0,$$

one can introduce the logarithmic gap

$$\Delta_n := \ln \frac{l}{q}.$$

Then the algebraic comparison of the two normalizations is written as

$$\Delta_n = \frac{1}{n} \ln \frac{2^{n-1}}{n}.$$

The key point of version 16.4 is that the zero logarithmic gap condition

$$\Delta_n = 0$$

is equivalent to the equality of residual scales

$$l^n = q^n$$

and, given the two specified normalizations of the same core, to the equality of coefficient masses

$$n = 2^{n-1}.$$

In other words, the central chain of version 16.4 takes the form

$$l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1}.$$

Next, Coq verifies an elementary final step:

$$n = 2^{n-1} \implies n \in \{1, 2\}.$$

Therefore, under the requirement of compatibility with a zero logarithmic gap, the exponent

$$n > 2$$

is excluded.

It is important to emphasize right away: this scheme remains conditional. Coq does not prove Fermat's Last Theorem unconditionally. It verifies the links between

the formalized premises, the zero logarithmic gap, the equality of residual scales, the equality of coefficient masses, and the final exponential-linear obstruction. The decisive mathematical point remains the question: must the internal two-power symmetry of a hypothetical positive integer solution generate the condition

$$\Delta_n = 0.$$

The entropic language, if used in the article, is only a heuristic interpretation of the logarithmic gap; no physical principle of entropy is used as an independent mathematical axiom or proof.

1. Correctness of Equalities for x^n and z^n

The equalities

$$x^n = (m^n - p^n)^n$$

and

$$z^n = (m^n + p^n)^n$$

are correct provided that it is preliminarily set

$$x = m^n - p^n,$$

$$z = m^n + p^n.$$

This is the usual raising of equal quantities to the same power.

In other words, if

$$x = m^n - p^n,$$

then automatically

$$x^n = (m^n - p^n)^n.$$

Similarly, if

$$z = m^n + p^n,$$

then automatically

$$z^n = (m^n + p^n)^n.$$

It is not required here that

$$m, \quad p$$

be integers. It is sufficient that the expressions be defined in the chosen numerical domain, for example in the domain of real numbers.

However, if it is assumed in advance that

$$x, z \in \mathbb{N},$$

then the equalities

$$x = m^n - p^n, \quad z = m^n + p^n$$

become consistency conditions between the real parameterization and the natural values

$$x, \quad z.$$

In the real case, there may be no arithmetic restrictions, but in the integer case, additional conditions of parity and exact power arise.

In Coq, this part is reflected by the lemma

`sum_diff_from_parameters_R,`

which captures the real identities: if

$$z = m^n + p^n, \quad x = m^n - p^n,$$

then

$$z + x = 2m^n, \quad z - x = 2p^n.$$

At the natural level, the lemma

`parameter_equations_raise_nat`

verifies an elementary fact: if

$$z = m^n + p^n, \quad x = m^n - p^n$$

are given, then after raising to the power, we automatically obtain

$$z^n = (m^n + p^n)^n, \quad x^n = (m^n - p^n)^n.$$

Thus, there is no hidden mathematical assumption at this step: only the elementary property of equality is used.

2. Real Factorization of the Odd Binomial Core

Consider the difference

$$(m^n + p^n)^n - (m^n - p^n)^n.$$

By the binomial formula, only odd binomial terms remain in it:

$$(m^n + p^n)^n - (m^n - p^n)^n = 2 \sum_{\substack{1 \leq j \leq n \\ j \text{ odd}}} \binom{n}{j} (m^n)^{n-j} (p^n)^j.$$

Let us denote the inner sum by

$$S_n(m, p) = \sum_{\substack{1 \leq j \leq n \\ j \text{ odd}}} \binom{n}{j} (m^n)^{n-j} (p^n)^j.$$

Then

$$y^n = 2S_n(m, p).$$

The first real normalization introduces a positive scale parameter

$$l$$

and writes the core in the form

$$S_n(m, p) = n l^n.$$

Equivalently, if

$$n > 0,$$

then

$$l^n = \frac{S_n(m, p)}{n}.$$

In this sense, the equality

$$y^n = 2n l^n$$

is correct as a real representation.

It is not required that

$$l$$

be an integer. If the positive real case is chosen, then it is assumed

$$l > 0.$$

In Coq, this is reflected by the definition

$$\text{real_core_factorization } n \text{ core } l,$$

which means

$$0 < n, \quad \text{core} = n l^n$$

over the real numbers. This is not a statement about integer divisibility, but exactly a real factorization.

The lemma

$$\text{real_core_factorization_substitutes_y_power}$$

shows: if

$$y^n = 2 \text{ core}$$

and

$$\text{core} = n l^n,$$

then

$$y^n = 2n l^n.$$

This corresponds exactly to the transition from the odd binomial core to the first scale notation.

In version 16.4, this block remains the algebraic foundation for the subsequent logarithmic layer. Logarithms appear only after the positive real residual factors

$$l > 0, \quad q > 0$$

have been introduced.

3. Difference Between Real Factorization and Integer Divisibility

If

$$l$$

is considered as a real number, there is no need to assert that the sum

$$S_n(m, p) = \sum_{\substack{1 \leq j \leq n \\ j \text{ odd}}} \binom{n}{j} (m^n)^{n-j} (p^n)^j$$

is divisible by

$$n$$

in integers.

One should strictly distinguish between two statements:

$$l^n = \frac{S_n(m, p)}{n}$$

as a real definition, and

$$n \mid S_n(m, p)$$

as an integer arithmetic statement.

The first is admissible in the real model. The second requires a separate proof and is not true in general.

For example, for

$$n = 6, \quad m = 1, \quad p = 1$$

the odd sum of binomial coefficients equals

$$\binom{6}{1} + \binom{6}{3} + \binom{6}{5} = 6 + 20 + 6 = 32.$$

But

$$32$$

is not divisible by

$$6$$

in integers.

This does not destroy the real scheme, because it does not assert the divisibility of the core by

$$n$$

in the ring

$$\mathbb{Z}.$$

The real scheme uses the notation

$$S_n(m, p) = n l^n$$

over

$$\mathbb{R}.$$

This is precisely why the definition

$$\text{real_core_factorization}$$

speaks of the representation of the core in the form

$$\text{core} = n l^n$$

over real numbers, and not that

$$n$$

is an integer divisor of the core.

This exact distinction is particularly important in version 16.4, since the logarithmic gap

$$\Delta_n = \ln \frac{l}{q}$$

is introduced over positive real residual factors

$$l > 0, \quad q > 0,$$

rather than over integer divisors.

4. Integer Specialization and Parity Conditions

If it is additionally required that

$$m, p \in \mathbb{Z},$$

then it follows from

$$z = m^n + p^n, \quad x = m^n - p^n$$

that

$$z + x = 2m^n, \quad z - x = 2p^n.$$

Consequently,

$$z + x$$

and

$$z - x$$

must be even.

In the Coq language, this is expressed by the lemma

`sum_diff_from_parameters_Z`

and the corollary

`parity_condition_Z`.

If at least one of the parity conditions is violated, then integer parameters

$$m, p \in \mathbb{Z}$$

in such a form are impossible. This is reflected by the lemmas

`no_parameters_if_parity_violation`, `no_parameters_if_odd`.

For instance, Coq checks a specific case

$$z = 2, \quad x = 1, \quad n = 3.$$

Here

$$z + x = 3$$

is odd, so there are no integers

$$m, p$$

such that

$$2 = m^3 + p^3, \quad 1 = m^3 - p^3.$$

This is formalized by the lemma

`no_parameters_for_example`.

Thus Coq separates the general real level from integer specialization. Real reconstruction may be formally correct, but integer reconstruction requires additional arithmetic conditions.

5. Two Real Normalizations of the Odd Binomial Core

The central object of the current scheme is the odd binomial core

$$S_n(m, p) = \sum_{\substack{1 \leq j \leq n \\ j \text{ odd}}} \binom{n}{j} (m^n)^{n-j} (p^n)^j.$$

It arises from the equality

$$(m^n + p^n)^n - (m^n - p^n)^n = 2S_n(m, p).$$

The first normalization extracts the linear binomial factor

$$n = C_n^1$$

and writes

$$S_n(m, p) = n l^n.$$

In Coq this corresponds to

$$\text{real_core_factorization } n \text{ core } l.$$

This definition means the real factorization

$$\text{core} = n l^n,$$

and not a statement about integer divisibility.

The second normalization extracts the total mass of the odd binomial coefficients

$$2^{n-1}.$$

The classical identity takes the form

$$\sum_{\substack{1 \leq j \leq n \\ j \text{ odd}}} \binom{n}{j} = 2^{n-1}.$$

Therefore, one can introduce a second positive real residual factor

$$q > 0$$

and write

$$S_n(m, p) = 2^{n-1} q^n.$$

Here

$$q$$

is a real residual factor. It should not be confused with a modulus in modular arithmetic.

In Coq, the second normalization corresponds to the definition

$$\text{coefficient_mass_factorization } n \text{ core } q,$$

which means

$$0 < n, \quad \text{core} = 2^{n-1} q^n$$

over the real numbers.

If the same core is written in two ways,

$$S_n(m, p) = n l^n, \quad S_n(m, p) = 2^{n-1} q^n,$$

then

$$n l^n = 2^{n-1} q^n.$$

After dividing by

$$n > 0$$

we obtain the exact relation

$$l^n = q^n \frac{2^{n-1}}{n}.$$

In Coq this is verified by the lemma

`two_scale_factorizations_relation`.

Its meaning consists of the following: from two real factorizations of the same core, it automatically follows that

$$l^n = q^n \frac{2^{n-1}}{n}.$$

This exact relation is the algebraic bridge to the logic of version 16.4. In version 16.4, it is additionally written in logarithmic form via

$$\Delta_n = \ln \frac{l}{q}.$$

Logarithmic Form of the Comparison

Since

$$l > 0, \quad q > 0,$$

one can take logarithms in the relation

$$l^n = q^n \frac{2^{n-1}}{n}.$$

We obtain

$$n \ln l = n \ln q + (n-1) \ln 2 - \ln n.$$

Consequently,

$$\ln l - \ln q = \frac{1}{n} \ln \frac{2^{n-1}}{n},$$

that is

$$\ln \frac{l}{q} = \frac{1}{n} \ln \frac{2^{n-1}}{n}.$$

Let us define

$$\Delta_n := \ln \frac{l}{q}.$$

Then

$$\Delta_n = \frac{1}{n} \ln \frac{2^{n-1}}{n}.$$

The quantity

$$\Delta_n$$

measures the logarithmic imbalance between the two residual factors. It can be heuristically interpreted as an entropy-type gap between two normalizations of the same odd binomial core. Strictly mathematically, however, this is only a logarithmic reformulation of the exact algebraic relation

$$l^n = q^n \frac{2^{n-1}}{n}.$$

No physical entropy principle is used here as an independent proof.

6. Logarithmic Obstruction to Coefficient Symmetry Compatibility

In version 16.4, the central structural point is not a one-sided hypothesis, but an exact criterion for the compatibility of two residual factors in logarithmic form. Simple equality of residual scales in Coq is defined as

$$\text{residual_scale_equality } n \ l \ q.$$

This means exactly

$$l^n = q^n.$$

The logarithmic gap in Coq is defined as

$$\text{logarithmic_gap } l \ q.$$

Mathematically this means

$$\ln \frac{l}{q}.$$

The zero logarithmic gap condition in Coq is defined as

$$\text{zero_logarithmic_gap } l \ q.$$

It means

$$\ln \frac{l}{q} = 0.$$

The equality of coefficient masses in Coq is defined as

$$\text{coefficient_mass_equality } n.$$

It means

$$n = 2^{n-1}.$$

Comparing the two normalizations yields

$$l^n = q^n \frac{2^{n-1}}{n}.$$

If the zero logarithmic gap condition is satisfied

$$\Delta_n = 0,$$

then, since

$$l > 0, \quad q > 0,$$

we have

$$\ln \frac{l}{q} = 0 \iff \frac{l}{q} = 1 \iff l = q.$$

This implies

$$l^n = q^n.$$

Conversely, if

$$l^n = q^n$$

and

$$l > 0, \quad q > 0,$$

then

$$l = q,$$

which means

$$\Delta_n = \ln \frac{l}{q} = 0.$$

Therefore

$$l^n = q^n \iff \Delta_n = 0.$$

Furthermore, given the two real normalizations of the same odd binomial core, from the exact relation

$$l^n = q^n \frac{2^{n-1}}{n}$$

it follows that

$$\Delta_n = 0 \iff n = 2^{n-1}.$$

Thus, the central chain of version 16.4 has the form

$$l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1}.$$

In Coq, the bridge from the equality of residual scales to the zero logarithmic gap is expressed by the lemma

`residual_scale_equality_implies_zero_logarithmic_gap.`

The reverse bridge is expressed by the lemma

`zero_logarithmic_gap_implies_residual_scale_equality.`

Both directions are collected into the theorem

`residual_scale_equality_iff_zero_logarithmic_gap.`

The former algebraic core is also preserved. The theorem

`coefficient_symmetry_compatibility_iff`

still expresses the equivalence

$$l^n = q^n \iff n = 2^{n-1}$$

given two real factorizations of the same core.

The new theorem of version 16.4

`logarithmic_coefficient_symmetry_compatibility_iff`

asserts that under the same factorizations and the positivity conditions

$$l > 0, \quad q > 0$$

the zero logarithmic gap is equivalent to the equality of coefficient masses:

$$\Delta_n = 0 \iff n = 2^{n-1}.$$

The theorem

`logarithmic_zero_gap_chain`

gathers both bridges into a single chain:

$$(l^n = q^n \iff \Delta_n = 0)$$

and

$$(\Delta_n = 0 \iff n = 2^{n-1}).$$

For convenience, Coq also introduces the structure

`LogarithmicCoefficientSymmetryData`.

This structure packages the data:

$$n, \quad \text{core}, \quad l, \quad q,$$

the positivity

$$l > 0, \quad q > 0,$$

the two factorizations

$$\text{core} = n l^n, \quad \text{core} = 2^{n-1} q^n,$$

and the zero logarithmic gap condition

$$\Delta_n = 0.$$

The lemma

`logarithmic_data_forces_shift`

derives from this structure

$$n = 2^{n-1}.$$

The lemma

`logarithmic_data_forces_small_exponent`

derives

$$n = 1 \quad \text{or} \quad n = 2.$$

The theorem

`logarithmic_data_excludes_high_exponent`

excludes the case

$$n > 2$$

for data satisfying the zero logarithmic gap.

7. Elementary Exponential-Linear Obstruction

It remains to analyze the equation

$$n = 2^{n-1}.$$

It is equivalent to

$$2^n = 2n.$$

In positive natural numbers, this equation has exactly two solutions:

$$n = 1, \quad n = 2.$$

Indeed,

$$1 = 2^{1-1} = 1,$$

$$2 = 2^{2-1} = 2.$$

And for any integer

$$n > 2$$

the exponential term

$$2^{n-1}$$

is strictly greater than the linear term

$$n.$$

In Coq, the corresponding part is broken down into several elementary lemmas:

pow2_gt_linear_shift,
pow2_gt_linear,
pow_eq_linear_positive,
two_pow_eq_two_mul_iff_shift,
binary_scaling_roots_only_one_two.

The final lemma

binary_scaling_roots_only_one_two

asserts:

$$n = 2^{n-1} \implies n = 1 \text{ or } n = 2.$$

In version 16.4, this exponential-linear obstruction appears as a consequence of the zero logarithmic gap:

$$\Delta_n = 0 \implies n = 2^{n-1} \implies n \in \{1, 2\}.$$

At the level of the packaged structure

LogarithmicCoefficientSymmetryData

this yields the lemma

logarithmic_data_forces_small_exponent,

which asserts:

$$\text{LogarithmicCoefficientSymmetryData} \implies n = 1 \text{ or } n = 2.$$

Finally, the theorem

$$\text{logarithmic_data_excludes_high_exponent}$$

states:

$$\text{LogarithmicCoefficientSymmetryData} \implies n > 2 \implies \text{False}.$$

The more general theorem

$$\text{logarithmic_zero_gap_excludes_high_exponent}$$

formalizes the final conditional theorem of version 16.4 directly from the following premises:

$$\begin{aligned} \text{core} &= n l^n, \\ \text{core} &= 2^{n-1} q^n, \\ l &> 0, \quad q > 0, \\ \Delta_n &= 0, \\ n &> 2. \end{aligned}$$

From these conditions, Coq derives a contradiction. That is, given two real factorizations of the same core and requiring a zero logarithmic gap, an exponent strictly greater than two is excluded.

8. Modular Remark

The present scheme is not a proof via modular arithmetic and does not assert the absence of solutions to corresponding congruences over finite fields.

The scheme under consideration applies to equalities over

$$\mathbb{N}, \quad \mathbb{Z}, \quad \mathbb{Q}, \quad \mathbb{R},$$

and not to equations over a finite field

$$\mathbb{F}_{\text{mod } q}.$$

From the integer equality

$$x^n + y^n = z^n$$

a congruence modulo any

$$\text{mod } q$$

indeed follows:

$$x^n + y^n \equiv z^n \pmod{\text{mod } q}.$$

However, the reverse transition is not true in general. From the congruence

$$x^n + y^n \equiv z^n \pmod{\text{mod } q}$$

the equality

$$x^n + y^n = z^n$$

in integers does not follow.

In other words, a modular congruence merely means that there exists an integer

$$t$$

such that

$$z^n = x^n + y^n + \text{modq} \cdot t.$$

An integer solution to Fermat's equation corresponds to the special case

$$t = 0.$$

Therefore, the existence of solutions for the congruences

$$x^n + y^n \equiv z^n \pmod{\text{modq}}$$

for

$$n > 2$$

is not a counterexample to Fermat's Last Theorem and does not refute the real parameterization used in this scheme.

If one specifically moves to the modular level, the parameters should be defined separately as residues. For example, instead of the real equalities

$$z = m^n + p^n, \quad x = m^n - p^n$$

one must separately fix

$$m^n \equiv A \pmod{\text{modq}}, \quad p^n \equiv B \pmod{\text{modq}}.$$

After that, the corresponding modular relations take the form

$$z \equiv A + B \pmod{\text{modq}}, \quad x \equiv A - B \pmod{\text{modq}}.$$

Thus, real parameterization and modular arithmetic belong to different levels of description.

In version 16.4, this modular remark retains its previous meaning: it does not directly participate in the logarithmic zero-gap obstruction, but it protects the reasoning from confounding different levels: integer equalities and modular congruences.

9. How the Modular Remark is Reflected in Coq

In Coq, there is a section

ModularRemark.

In it, the relation of modular congruence is defined as

$$\text{modular_congruent modq } a \ b.$$

Mathematically this means that there exists an integer

$$t$$

such that

$$a = b + \text{mod}q \cdot t.$$

The lemma

`integer_equality_implies_modular_power_congruence`

verifies the direction:

$$\text{integer equality} \implies \text{modular congruence}.$$

If

$$x^n + y^n = z^n,$$

then one can take

$$t = 0,$$

and obtain the corresponding congruence modulo any

$$\text{mod}q.$$

The lemma

`integer_equality_is_zero_modular_witness`

explicitly establishes that a true equality in integers corresponds to the zero modular witness

$$t = 0.$$

The lemma

`modular_congruence_not_integer_equality`

verifies a concrete example:

$$3^3 \equiv 1^3 + 1^3 \pmod{5},$$

since

$$27 = 2 + 5 \cdot 5,$$

but at the same time

$$1^3 + 1^3 \neq 3^3$$

as an equality in integers, since

$$2 \neq 27.$$

The structure

`ModularResidues`

formalizes the idea that at the modular level, residues must be specified separately

$$m^n \equiv A \pmod{\text{mod}q}, \quad p^n \equiv B \pmod{\text{mod}q},$$

and then separately one must set

$$z \equiv A + B \pmod{\text{mod}q}, \quad x \equiv A - B \pmod{\text{mod}q}.$$

Thus, Coq strictly confirms the methodological idea:

$$\text{integer equality} \implies \text{modular congruence},$$

but

$$\text{modular congruence} \not\Rightarrow \text{integer equality}.$$

10. Overall Structure of the Coq Verification

The Coq file of version 16.4 verifies a conditional algebraic-logarithmic chain. It can be written as follows:

$$\begin{aligned}\text{core} &= n l^n, \\ \text{core} &= 2^{n-1} q^n.\end{aligned}$$

From these two real factorizations of the same odd binomial core follows the exact relation

$$l^n = q^n \frac{2^{n-1}}{n}.$$

Next, for

$$l > 0, \quad q > 0,$$

the logarithmic gap is introduced

$$\Delta_n = \ln \frac{l}{q}.$$

Coq verifies the bridges:

$$\begin{aligned}l^n = q^n &\iff \Delta_n = 0, \\ \Delta_n = 0 &\iff n = 2^{n-1}, \\ n = 2^{n-1} &\implies n = 1 \text{ or } n = 2.\end{aligned}$$

The final conclusion is as follows: with

$$\text{zero_logarithmic_gap}$$

and

$$n > 2$$

a contradiction arises.

The main Coq objects have the following meaning:

- `real_core_factorization` means

$$\text{core} = n l^n$$

over the real numbers.

- `coefficient_mass_factorization` means

$$\text{core} = 2^{n-1} q^n$$

over the real numbers.

- `two_scale_factorizations_relation` derives from two factorizations

$$l^n = q^n \frac{2^{n-1}}{n}.$$

- `residual_scale_equality` means

$$l^n = q^n.$$

- `coefficient_mass_equality` means

$$n = 2^{n-1}.$$

- `logarithmic_gap` means

$$\ln \frac{l}{q}.$$

- `zero_logarithmic_gap` means

$$\ln \frac{l}{q} = 0.$$

- `coefficient_logarithmic_gap` specifies the coefficient side of the logarithmic gap:

$$\frac{1}{n} \ln \frac{2^{n-1}}{n}.$$

- `residual_scale_equality_implies_zero_logarithmic_gap` derives

$$\Delta_n = 0$$

from

$$l^n = q^n$$

given the positivity

$$l > 0, \quad q > 0.$$

- `zero_logarithmic_gap_implies_residual_scale_equality` derives

$$l^n = q^n$$

from

$$\Delta_n = 0$$

given the positivity

$$l > 0, \quad q > 0.$$

- `residual_scale_equality_iff_zero_logarithmic_gap` collects both directions:

$$l^n = q^n \iff \Delta_n = 0.$$

- `coefficient_symmetry_compatibility_iff` preserves the previous algebraic core:

$$l^n = q^n \iff n = 2^{n-1}.$$

- `logarithmic_coefficient_symmetry_compatibility_iff` expresses the new logarithmic formulation of version 16.4:

$$\Delta_n = 0 \iff n = 2^{n-1}.$$

- `logarithmic_zero_gap_chain` formalizes the full chain:

$$l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1}.$$

- `CoefficientSymmetryData` preserves the old data packaging via the equality of residual scales.

- `LogarithmicCoefficientSymmetryData` introduces the new data packaging of version 16.4 via the zero logarithmic gap.
- `logarithmic_data_forces_shift` derives from this structure

$$n = 2^{n-1}.$$

- `binary_scaling_roots_only_one_two` verifies

$$n = 2^{n-1} \implies n = 1 \text{ or } n = 2.$$

- `logarithmic_data_forces_small_exponent` derives from the logarithmic structure

$$n = 1 \text{ or } n = 2.$$

- `logarithmic_data_excludes_high_exponent` derives from the logarithmic structure

$$n > 2 \implies \text{False}.$$

- `logarithmic_zero_gap_excludes_high_exponent` derives the impossibility of the case

$$n > 2$$

directly from the two factorizations, the positivity

$$l > 0, \quad q > 0,$$

and the condition

$$\Delta_n = 0.$$

In other words, version 16.4 does not discard the previous algebraic logic. It adds a logarithmic layer to it:

$$l^n = q^n \iff \Delta_n = 0.$$

Therefore, the former obstacle of coefficient symmetry compatibility turns into the logarithmic obstruction of compatibility with a zero gap.

11. What Coq Does Not Prove

For fairness, it should be explicitly stated that the Coq file does not prove Fermat's Last Theorem unconditionally.

It does not automatically prove that from every hypothetical positive integer solution of the equation

$$x^n + y^n = z^n$$

for

$$n > 2$$

the zero logarithmic gap condition

$$\Delta_n = 0$$

necessarily follows.

Coq verifies another, strictly conditional statement. If two real factorizations of the same odd binomial core

$$\text{core} = n l^n, \quad \text{core} = 2^{n-1} q^n,$$

are given, if

$$l > 0, \quad q > 0,$$

and if additionally the condition

$$\Delta_n = 0$$

is satisfied, then Coq derives

$$n = 2^{n-1},$$

and then

$$n \in \{1, 2\}.$$

Consequently, under these premises, the exponent

$$n > 2$$

is impossible.

Thus, the weak point of the scheme is not in the real parameterization, not in the notation

$$\text{core} = n l^n,$$

not in the modular remark, and not in the Coq-verified final exponential-linear step. The decisive mathematical point of version 16.4 remains the status of the requirement for compatibility with a zero logarithmic gap:

$$\Delta_n = 0.$$

It is exactly this point that should be the subject of external mathematical critique.

12. Alignment with the Current Article

In the current article of version 16.4, the main conditional result is formulated as follows. If a hypothetical positive natural solution to Fermat's equation

$$x^n + y^n = z^n, \quad n > 2,$$

admits the symmetric real parameterization under consideration, and if the same odd binomial core admits two natural real normalizations

$$\text{core} = n l^n, \quad \text{core} = 2^{n-1} q^n,$$

then their comparison yields

$$l^n = q^n \frac{2^{n-1}}{n}.$$

For

$$l > 0, \quad q > 0$$

this can be written via the logarithmic gap

$$\Delta_n = \ln \frac{l}{q}.$$

Then

$$l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1}.$$

Since

$$n = 2^{n-1}$$

is possible only for

$$n = 1 \quad \text{or} \quad n = 2,$$

then under the requirement of compatibility with a zero logarithmic gap, the exponent

$$n > 2$$

is excluded.

In this sense, the Coq file `GlobalNormalization.v` of version 16.4 matches the article: it verifies exactly the formal conditional chain, and not an unconditional Fermat's Last Theorem.

The filename `GlobalNormalization.v` is kept for backward compatibility of the GitHub repository; the mathematical content of version 16.4 corresponds to the logarithmic *zero-gap obstruction*.

The modular remark remains a separate methodological block. It does not participate in the proof of the final exponential-linear obstruction, but it protects the reasoning from confounding different levels: integer equalities and modular congruences.

Conditional Theorem. Let the same odd binomial core admit two real factorizations for some natural exponent

$$n > 0$$

$$\text{core} = n l^n, \quad \text{core} = 2^{n-1} q^n,$$

where

$$l > 0, \quad q > 0.$$

Then their comparison yields

$$l^n = q^n \frac{2^{n-1}}{n}.$$

If additionally the zero logarithmic gap condition is satisfied

$$\Delta_n = \ln \frac{l}{q} = 0,$$

then

$$l^n = q^n$$

and, consequently,

$$n = 2^{n-1}.$$

Since

$$n = 2^{n-1} \implies n = 1 \text{ or } n = 2,$$

then under the requirement of a zero logarithmic gap, the case

$$n > 2$$

is impossible.

Final Conclusion. The final Coq verification of version 16.4 has the following exact status.

1. Real parameterization is verified as an elementary algebraic reconstruction.
2. Integer specialization is separated from the real level.
3. Modular congruences are separated from integer equalities.
4. Two real normalizations of the same odd binomial core yield an exact relation between

$$l \quad \text{and} \quad q.$$

5. For

$$l > 0, \quad q > 0$$

this relation admits a logarithmic representation via

$$\Delta_n = \ln \frac{l}{q}.$$

6. The condition

$$\Delta_n = 0$$

is equivalent to

$$l^n = q^n.$$

7. Given two normalizations, this is equivalent to

$$n = 2^{n-1}.$$

8. Coq verifies that

$$n = 2^{n-1}$$

implies

$$n \in \{1, 2\}.$$

9. Therefore, under the requirement of a zero logarithmic gap, the case

$$n > 2$$

is impossible.

10. But Coq does not prove that every hypothetical positive integer solution to Fermat's equation must yield

$$\Delta_n = 0.$$

11. Therefore, version 16.4 remains a conditional reconstruction, and not an unconditional proof of Fermat's Last Theorem.

The main Coq-verified chain has the form

$$\begin{aligned}
& \text{core} = n l^n, \quad \text{core} = 2^{n-1} q^n \\
& \implies \\
& l^n = q^n \frac{2^{n-1}}{n} \\
& \implies \\
& l^n = q^n \iff \Delta_n = 0 \iff n = 2^{n-1} \\
& \implies \\
& n \in \{1, 2\}.
\end{aligned}$$

The entropic language of version 16.4 should be understood only as a heuristic interpretation of the logarithmic gap. The strict mathematical content consists of the algebraic-logarithmic equivalence and the conditional Coq-verified exclusion of exponents

$$n > 2$$

when

$$\Delta_n = 0$$

holds.